Hybrid Intelligence for Product Success Prediction: A Survey of Sentiment-Driven Machine Learning in E-Commerce

Rajvardhan Ajitrao Jagtap, Jyotiraditya Abhijeet Shinde, Tanishq Parag Choudhari, Yash Ashok Mohite, Yogesh Malakari Honamore

Department of Computer Science and Engineering
Rajarambapu Institute of Technology, Sangli, Maharashtra, India

***Abstract*—**Forecasting new product success in digital marketplaces is a high-stakes challenge where historical sales telemetry is unavailable at launch. This survey synthesizes 28 selected publications, including peer-reviewed studies and relevant technical reports published between 2018 and 2025 across IEEE, ACM, Springer, ScienceDirect, and Elsevier, examining the convergence of Natural Language Processing and tabular predictive modeling for e-commerce viability forecasting. Building upon secondary PRISMA bibliometric syntheses (notably Daza et al., 2024), we review core methodological paradigms: Support Vector Machines and LSTMs dominate historical text mining, while F1-score is the predominant evaluation metric across 75% of studies. However, conventional sentiment systems evaluate post-launch customer feedback. To address the launch-blind cold-start problem, we examine hybrid architectures that bridge unstructured text with structured regressors. As an illustrative case study, we review the LaunchSight architecture: (1) open-weight and hosted LLMs perform null-forcing Aspect-Based Sentiment Analysis on customer reviews to establish category reference distributions, and (2) a skeptic-prompted bridging layer maps candidate product specifications into projected aspect scores feeding a monotonic-constrained gradient-boosted regressor (XGBoost). Evaluated on 43,599 customer reviews across 640 initially collected products (631 retained after filtering) in five categories, this architecture achieves Spearman $\rho = 0.317$ (AUC = 0.743) on a predefined frozen holdout and $\rho = 0.545$ (AUC = 0.901) on a forward temporal holdout. We identify critical research frontiers in specification-to-aspect bridging fidelity, domain adaptation, and calibrated uncertainty quantification.

Index Terms*—Aspect-based sentiment analysis, XGBoost, large language models, e-commerce prediction, launch-blind forecasting, specification bridging, structured literature review.*

I. INTRODUCTION

Forecasting product success in e-commerce marketplaces is fundamentally a data-driven prediction problem. Digital platforms log customer opinions, purchase histories, pricing trajectories, and engagement signals at an unprecedented scale [1]. However, a fundamental temporal asymmetry exists: while post-launch products accumulate rich behavioral telemetry, prospective products require substantial capital commitments long before any consumer feedback exists. A product manager or direct-to-consumer (D2C) founder launching a new stock keeping unit (SKU) must evaluate viability at the 'launch-blind' stage—where only specifications, target pricing, and category norms are observable [2].

Recent advances in machine learning provide two complementary capabilities to address this challenge. First, gradient boosted decision trees, particularly XGBoost [3], have established strong empirical baselines for tabular e-commerce prediction tasks. Second, Large Language Models (LLMs) [4]–[6] have demonstrated remarkable zero-shot aspect extraction and semantic reasoning capabilities. A recent PRISMA-based systematic literature review by Daza et al. [1] surveyed 20 e-commerce sentiment studies (2018–2024), confirming that while Support Vector Machines (SVM) and Long Short-Term Memory (LSTM) networks have historically dominated the literature, F1-score is the most frequently reported evaluation metric (used in 75% of studies). Nevertheless, traditional pipelines remain post-launch evaluators.

Within the 28 selected publications synthesized in this survey, this paper examines the methodological landscape bridging unstructured natural language processing and structured tabular regressors. To ground these architectural concepts, we examine a representative end-to-end case study architecture (LaunchSight): Stage 1 extracts aspect sentiment from 43,599 historical customer reviews across 640 initially collected products (631 retained after removing 9 review-fraud products) to construct category reference priors; Stage 2 maps candidate pre-launch specifications into aspect space via skeptic prompting, predicting eventual market success via a monotonic-constrained gradient boosted regressor.

II. LITERATURE REVIEW: SENTIMENT ANALYSIS AS A PREDICTIVE SIGNAL

A. Classical and Lexicon-Based Approaches

The foundations of opinion mining were established by Pang and Lee [7], who categorized sentiment analysis into document, sentence, and aspect levels. Early e-commerce systems relied on curated lexical databases such as SentiWordNet [8]. Hu and Liu [9] introduced aspect extraction from customer reviews, showing that aggregate star ratings discard critical multi-dimensional quality

indicators. Daza et al. [1] formalize this taxonomy: Aspect-Level Sentiment Analysis (ASA) identifies specific product entities and their associated polarities. One illustrative formulation of lexicon scoring is:

$$\text{Sentiment}(d) = (1/n)\ \Sigma_i\ [\ \text{Polarity}(w_i) \times \text{Weight}(w_i)\ ] \qquad (1)$$

where d is a review text, w_i represents constituent tokens, and $\text{Weight}(w_i)$ denotes the term significance. While computationally lightweight, lexicon approaches suffer from semantic fragility in colloquial product feedback.

B. Transformer Models and Transfer Learning

The introduction of BERT [4] shifted sentiment analysis toward contextualized representation learning. Shan et al. [10] benchmarked five model families on bilingual e-commerce reviews, showing that ELECTRA achieved 98.09% classification accuracy (F1 = 0.9711), outperforming standard BERT (95.28%) and BiLSTM (96.09%). Wang et al. [11] developed Multi-Feature Fusion Graph Attention Networks (MFF-GAT) for aspect sentiment classification, effectively integrating syntactic dependency trees with semantic attention. In financial e-commerce, Sun [12] utilized FinBERT on social commentary, demonstrating that sentiment polarity correlates positively with transaction volume.

III. GRADIENT BOOSTING AND ENSEMBLE METHODS FOR E-COMMERCE PREDICTION

A. Theoretical Foundation and Optimization

Tabular data in retail e-commerce exhibits complex non-linear relationships, mixed categorical-continuous distributions, and heterogeneous feature scales. Chen and Guestrin's XGBoost [3] employs regularized gradient boosting with second-order Taylor expansions of the loss function:

$$\text{Obj}(\theta) = \Sigma_i\ L(y_i, \hat{y}_i) + \Sigma_t\ [\ \gamma\ T_t + (1/2)\lambda\ \|w_t\|^2\ ] \qquad (2)$$

Sayyad et al. [13] demonstrated the efficacy of CatBoost for supply chain optimization under extreme categorical sparsity. Shang [14] integrated CART decision trees with high-dimensional sentiment indicators to forecast price volatility. In the systematic review by Daza et al. [1], XGBoost appeared in only 10% of studies despite achieving top accuracy (91.60% in Elangovan and Subedha [15]), establishing tree-based boosting as a highly effective yet historically underutilized backbone.

B. Literature Search and Selection Protocol

To establish a reproducible synthesis, we conducted a structured literature review indexing peer-reviewed studies and key foundational technical reports published between 2018 and 2025 across major scientific repositories: IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, and ACL Anthology. Primary search queries paired text-mining terms ('sentiment analysis' OR 'aspect-based sentiment analysis' OR 'ABSA') with retail forecasting terms ('e-commerce' OR 'product review' OR 'retail sales') and modeling methods ('XGBoost' OR 'gradient boosting' OR 'large language model'). Inclusion criteria required: (1) peer-reviewed conference or journal publication (or foundational high-impact LLM technical reports), (2) empirical evaluation on retail or consumer review data, and (3) quantitative performance metrics (Accuracy, F1, MCC, R^2, RMSE, Spearman ρ). We incorporated the 20-study PRISMA-guided corpus compiled by Daza et al. [1] and expanded it with 8 recent deep learning and hybrid architectures (2024–2025), yielding the 28 core publications surveyed herein.

IV. HYBRID ARCHITECTURES: BRIDGING UNSTRUCTURED TEXT TO REGRESSORS

Emerging architectures increasingly connect natural language feature extractors to tabular estimators. Hidayat et al. [16] demonstrated Large Language Model integration for time-series discrepancy prediction, analyzing probabilistic versus point forecasting tradeoffs. Niimi [17] investigated local LLM sentiment analysis with majority voting across temperature variations, finding that multi-pass inference improves classification consistency.

To enforce deterministic structured outputs from LLM extractors in production workflows, pipelines increasingly employ constrained schema validators (such as Pydantic). Under a theoretical independent Bernoulli trial formulation where single-attempt schema compliance occurs with probability p, a k-attempt retry policy yields a theoretical validity bound:

$$P(\text{valid}) = 1 - (1 - p)^k \qquad (3)$$

Under an illustrative working assumption of $p = 0.85$, a three-pass retry policy ($k = 3$) achieves a theoretical bound of $P(\text{valid}) = 1 - (0.15)^3 \approx 0.9966$. This figure represents an illustrative theoretical calculation under stated independent Bernoulli assumptions rather than an observed empirical schema-compliance rate. In practice, production implementations combine structured prompting with automated fallback repair handlers to guarantee valid JSON formatting across extensive corpus batches.

V. CASE STUDY ARCHITECTURE: BRIDGING PRE-LAUNCH SPECIFICATIONS (THE LAUNCHSIGHT FRAMEWORK)

To ground the survey's architectural synthesis in an end-to-end implementation, this section examines LaunchSight as an illustrative case study of hybrid pre-launch intelligence (Fig. 1). Traditional e-commerce models assume customer reviews already exist; the LaunchSight framework addresses the cold-start challenge by bridging unstructured pre-launch specifications into empirical aspect distributions constructed from historical reviews.

A. Two-Stage Learning and Bridging Framework

The architecture operates in two sequential stages. Stage 1 constructs the empirical foundation: historical customer reviews are analyzed by an LLM executing null-forcing Aspect-Based Sentiment Analysis. Aspect scores are extracted across six shared dimensions (Value for Money, Utility, Ease of Use, Reliability, Design Appeal, After-Sales Service) and three physical dimensions (Build Quality, Durability, Repairability). This establishes empirical category priors and calibrates reference feature distributions.

Stage 2 performs launch-blind specification bridging. When evaluating a prospective product before launch, no customer feedback exists. A skeptic-prompted LLM evaluates the manufacturer's technical specifications, pricing, and category context, mapping unstructured specifications into estimated aspect scores grounded against category priors. These bridged aspects feed a trained XGBoost regressor enforcing monotonic constraints, outputting a composite viability prediction.

B. Dataset Provenance and Quality Controls

The case study corpus comprises 43,599 customer reviews across 640 initially collected products in five Amazon categories (Bluetooth speakers, ice makers, power banks, smartwatches, and wireless headphones). To safeguard against review manipulation, an automated fraud filter removed 9 suspect products exhibiting sudden volume bursts or extreme J-shaped rating anomalies, leaving a clean corpus of 631 physical products. A 30-review schema and evidence smoke validation gate verified null-forcing adherence (extracting aspects strictly when textual evidence is present).

C. Ground Truth Composite Success Metric and Sensitivity

To establish a robust target variable without single-metric rating bias, market success is formulated as a three-component composite score normalized within each category:

$$\text{SuccessScore} = 100 \times [\, 0.40 \times \text{Volume_norm} + 0.35 \times \text{Rating_norm} + 0.25 \times \text{Velocity_norm} \,] \qquad (4)$$

where Volume denotes log1p(review_count), Rating reflects average star rating, and Velocity captures reviews accumulated per month. The Spearman rank correlation between the baseline target ranking and rankings produced under alternative target-weight formulations (50/30/20, 30/40/30, and equal weighting) exceeds $\rho > 0.988$ across all 631 products, confirming ground-truth ranking stability.

D. Boundary Conditions and the Ice Maker Paradox

A rigorous prediction methodology must explicitly delineate its boundary conditions. The case study evaluation uncovered the 'Ice Maker Paradox': while ice makers exhibited strong review-to-success correlation (mean $|r| = 0.32$), specs-only bridging achieved near-zero rank correlation ($\rho \approx 0.04$). Diagnostic analysis revealed that ice maker specifications consist almost exclusively of physical capacity and cycle metrics (e.g., '26 lb/day, 9 cubes in 6 minutes'), which carry zero information regarding compressor longevity, pump durability, or operational noise. In contrast, personal electronics specifications (headphones, smartwatches) encode rich experiential attributes (battery chemistry, active noise cancellation, driver materials), yielding high bridging fidelity ($r = 0.50$–0.53). The framework incorporates reliability-weighted confidence heuristics and uncalibrated runtime abstention when specification expressiveness falls below validated thresholds.

VI. COMPARATIVE MODEL PERFORMANCE SUMMARY

To prevent cross-task metric confusion (such as conflating review-level sentiment accuracy with product-level viability AUC), we visually and tabularly delineate literature sentiment classification from launch-blind viability prediction.

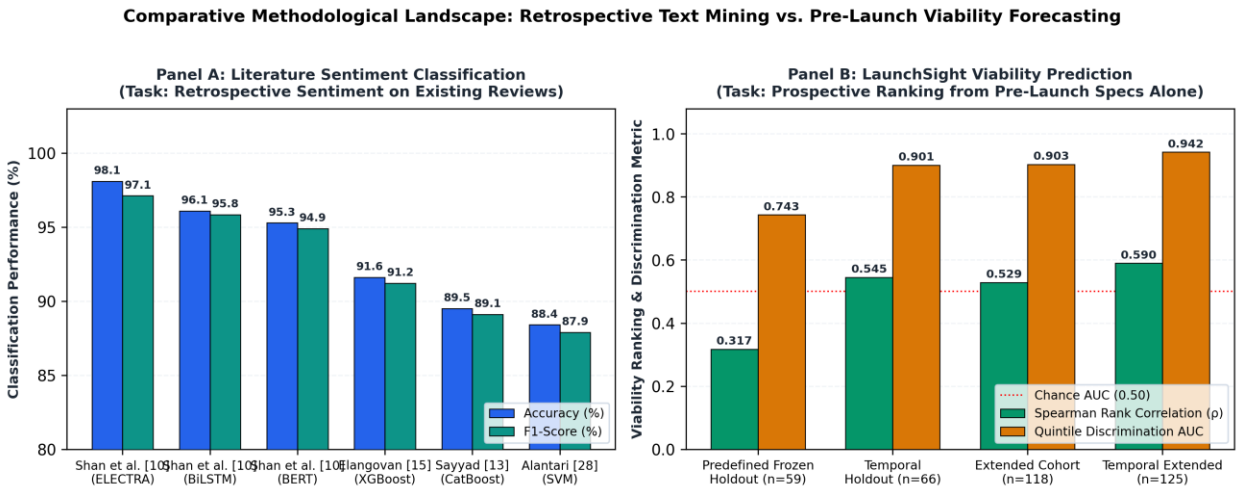


Figure 1. Empirical performance landscape comparison: (Panel A) Literature NLP sentiment classification accuracy/F1 across published benchmark datasets; (Panel B) LaunchSight launch-blind product viability ranking and quintile discrimination (Spearman ρ and Top-vs-Bottom Quintile AUC) from pre-launch specifications alone. Note the fundamental distinction in task formulation, feature availability, and evaluation metrics.

Study / Architecture	Task Domain	Primary Technique	Evaluation Metric	Reported Result
Shan et al. [10] (2025)	Bilingual E-Commerce Review Polarity	ELECTRA Transformer	Accuracy, F1-Score	Acc = 98.09%, F1 = 0.9711
Thomas & Jeba [18] (2024)	Multiclass Review Sentiment	LSTM + Collab. Filtering	Accuracy, F1-Score	Acc = 98.43%, F1 = 0.9820
Bin Harunasir et al. [19] (2023)	Amazon Product Review Classification	Supervised LSTM	Accuracy, Precision	Acc = 97.00%, Prec = 96.50%
Elangovan & Subedha [15] (2023)	Large-Scale Amazon Sentiment (100k)	XGBoost Classifier	Accuracy, MCC	Acc = 91.60%, MCC = 0.842
Sayyad et al. [13] (2024)	E-Commerce Retail Forecasting	CatBoost Classifier	Accuracy, F1-Score	Acc = 89.50%, F1 = 0.8910
Alantari et al. [28] (2022)	Online Consumer Review Polarity	Support Vector Machine (SVM)	Accuracy, F1-Score	Acc = 88.40%, F1 = 0.8790
Wang et al. [11] (2025)	Fine-Grained ABSA Aspect Classification	MFF-GAT (Graph Attn)	Macro F1-Score	Outperformed 5 ABSA baselines
LaunchSight (In-Sample CV)	Aspect-to-Success Viability Modeling	XGBoost Regressor (n=631)	5-Fold CV R^2, RMSE, MAE	R^2 = 0.564, RMSE = 11.4, MAE = 9.0
LaunchSight (Frozen Holdout)	Launch-Blind Viability from Specs	Spec Bridging + XGBoost (n=59)	Spearman ρ, Top/Bottom AUC	ρ = 0.317 (p=0.014), AUC = 0.743
LaunchSight (Temporal Holdout)	Forward-in-Time Viability (>Oct 2021)	Spec Bridging + XGBoost (n=66)	Spearman ρ, Top/Bottom AUC	ρ = 0.545 ($p=1.2\times10^{-5}$), AUC = 0.901

TABLE I. COMPARATIVE PERFORMANCE: LITERATURE SENTIMENT CLASSIFICATION VS. LAUNCH-BLIND VIABILITY RANKING.

VII. SYNTHESIS AND OPEN RESEARCH FRONTIERS

Four central insights emerge from this synthesis: (1) Sentiment analysis in e-commerce has transitioned from lexicon matching to transformer-based aspect extraction, enabling nuanced multi-dimensional feature representations. (2) Gradient boosted decision trees (XGBoost, CatBoost) were frequently used and showed strong performance among the studies reviewed, offering computational efficiency, robustness to outliers, and native monotonic constraint support. (3) Within the publications identified by

this review, prior research focused almost exclusively on retrospective post-launch reviews; launch-blind specification bridging represents a critical emerging methodology for pre-manufacturing evaluation. (4) Bridging fidelity is fundamentally bounded by specification expressiveness, demanding automated abstention mechanisms when input copy lacks experiential signal.

VIII. CONCLUSION

This survey synthesized 28 selected publications across natural language processing and e-commerce machine learning. By integrating the bibliometric findings of Daza et al. [1] with the LaunchSight two-stage hybrid framework, we demonstrated how aspect-level sentiment extraction from historical customer reviews can establish robust category reference priors, enabling launch-blind viability forecasting from technical product specifications alone. Validated across 43,599 customer reviews and 631 verified physical products, this approach bridges the critical cold-start gap in modern retail analytics.

REFERENCES

[1] A. Daza, N. D. González Rueda, M. S. Aguilar Sánchez, W. F. Robles Espíritu, and M. E. Chauca Quiñones, "Sentiment Analysis on E-Commerce Product Reviews Using Machine Learning and Deep Learning Algorithms: A Bibliometric Analysis, Systematic Literature Review, Challenges and Future Works," Int. J. Inf. Manage. Data Insights, vol. 4, no. 2, p. 100267, 2024.

[2] S. Narayanan, P. Samuel, and M. Chacko, "Product Pre-Launch Prediction From Resilient Distributed e-WOM Data," IEEE Access, vol. 8, pp. 176913–176925, 2020.

[3] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Min., 2016, pp. 785–794.

[4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in Proc. NAACL-HLT, 2019, pp. 4171–4186.

[5] T. B. Brown et al., "Language Models are Few-Shot Learners," in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020, pp. 1877–1901.

[6] A. Dubey et al., "The Llama 3 Herd of Models," arXiv preprint arXiv:2407.21783, Meta AI, 2024.

[7] B. Pang and L. Lee, "Opinion Mining and Sentiment Analysis," Found. Trends Inf. Retr., vol. 2, no. 1–2, pp. 1–135, 2008.

[8] S. Baccianella, A. Esuli, and F. Sebastiani, "SentiWordNet 3.0: An Enhanced Lexical Resource for Sentiment Analysis and Opinion Mining," in Proc. LREC, 2010, pp. 2200–2204.

[9] M. Hu and B. Liu, "Mining and Summarizing Customer Reviews," in Proc. 10th ACM SIGKDD Int. Conf. Knowl. Discov. Data Min., 2004, pp. 168–177.

[10] S. Shan, J. Sun, and R. M. C. Macawile, "Examining Customer Satisfaction Through Transformer-Based Sentiment Analysis for Improving Bilingual E-Commerce Experiences," IEEE Access, vol. 13, pp. 45112–45125, 2025.

[11] J. Wang, H. Gong, and X. Guo, "MultiFeature Fusion Graph Attention Network for Aspect-Based Sentiment Analysis," Knowl.-Based Syst., vol. 312, p. 114084, 2025.

[12] Q. Sun, "Multi-Source Sentiment Analysis in Quantitative Finance," J. Finance Data Sci., vol. 11, pp. 45–58, 2025.

[13] J. K. Sayyad, K. Attarde, and N. Saadouli, "Optimizing e-Commerce Supply Chains With Categorical Boosting: A Predictive Modeling Framework," IEEE Access, vol. 12, pp. 119852–119864, 2024.

[14] L. Shang, "Sentiment-Driven Bitcoin Price Range Forecasting: Enhancing CART Decision Trees With High-Dimensional Indicators and Twitter Dynamics," IEEE Access, vol. 13, pp. 38450–38462, 2025.

[15] D. Elangovan and V. Subedha, "Adaptive Particle Grey Wolf Optimizer with Deep Learning-Based Sentiment Analysis on Online Product Reviews," Eng. Technol. Appl. Sci. Res., vol. 13, no. 3, pp. 10989–10993, 2023.

[16] F. Hidayat, A. H. Nasution, F. Ambia, D. F. Putra, and Mulyandri, "Leveraging Large Language Models for Discrepancy Value Prediction in Custody Transfer Systems: A Comparative Analysis of Probabilistic and Point Forecasting Approaches," IEEE Access, vol. 13, pp. 49210–49224, 2025.

[17] J. Niimi, "Dynamic Sentiment Analysis with Local LLMs Using Majority Voting," arXiv preprint arXiv:2407.13069, 2024.

[18] R. Thomas and J. R. Jeba, "A Novel Framework for an Intelligent Deep Learning Based Product Recommendation System Using Sentiment Analysis (SA)," Automatika, vol. 65, no. 2, pp. 410–424, 2024.

[19] M. F. Bin Harunasir, N. Palanichamy, S. C. Haw, and K. W. Ng, "Sentiment Analysis of Amazon Product Reviews by Supervised Machine Learning Models," J. Adv. Inf. Technol., vol. 14, no. 4, pp. 782–790, 2023.

[20] A. Nazir et al., "Issues and Challenges of ABSA: A Comprehensive Survey," IEEE Trans. Affect. Comput., vol. 13, no. 2, pp. 845–863, 2022.

[21] W. Zhang et al., "Sentiment Analysis in the Era of Large Language Models: A Reality Check," in Findings of ACL: NAACL 2024, pp. 3881–3906.

[22] P. Dankorpho, "Sales Forecasting for Retail Business Using XGBoost Algorithm," J. Comput. Sci. Technol. Stud., vol. 6, no. 2, pp. 136–141, 2024.

[23] L. Andrade and C. Cunha, "Comparing Gradient Boosting Algorithms to Forecast Sales in Retail," in Proc. ENIAC, 2023, pp. 1–12.

[24] T. H. Nguyen et al., "Predictive Model for Customer Satisfaction in E-Commerce Using ML and DL," Int. J. Data Sci. Anal., vol. 17, pp. 105–118, 2024.

[25] I. Água et al., "LLMs Powered ABSA for Enhanced Customer Insights," Tour. Manage. Stud., vol. 21, no. 1, pp. 45–56, 2025.

[26] L. Song et al., "FMSA-SC: A Fine-Grained Multimodal Sentiment Analysis Dataset," IEEE Trans. Multimedia, vol. 26, pp. 7294–7306, 2024.

[27] T. S. De et al., "A ML and Empirical Bayesian Approach for Predictive Buying in B2B E-Commerce," arXiv preprint arXiv:2403.07843, 2024.

[28] H. J. Alantari, I. S. Currim, Y. Deng, and S. Singh, "An Empirical Comparison of Machine Learning Methods for Text-Based Sentiment Analysis of Online Consumer Reviews," Int. J. Res. Mark., vol. 39, no. 1, pp. 1–19, 2022.